Linear Algebra Selected Mother Problems

A focused review version based on Gilbert Strang, *Introduction to Linear Algebra*, Chapters 1–7

整理：Roxin-ChaI 项目重点版

使用方式

这份文档按“母题”标准重新筛选：不是为了覆盖所有小题，而是选择做透以后能迁移到同知识点其他题目的代表题。题目正文保留英文，不翻译；每道题都标明原文 Chapter、Subsection 和 Problem number。

- [Core]: first-pass mother problems; do these first.
- [Drill]: reinforcement problems; use them to confirm the same template works in nearby forms.
- [Challenge]: synthesis problems; use them to test whether the idea transfers across sections.

目录

1 Chapter 1: Introduction to Vectors

1. [Core]

Original: Chapter 1: Introduction to Vectors; Vectors and Linear Combinations; Problem 1.

母题覆盖：线性组合的几何判别：共线、共面、生成整个空间。

Describe geometrically (line, plane, or all of R^3) all linear combinations of

(a) $\begin{bmatrix}1\\2\\3\end{bmatrix}$ *and* $\begin{bmatrix}3\\6\\9\end{bmatrix}$ (b) $\begin{bmatrix}1\\0\\0\end{bmatrix}$ *and* $\begin{bmatrix}0\\2\\3\end{bmatrix}$ (c) $\begin{bmatrix}2\\0\\0\end{bmatrix}$ *and* $\begin{bmatrix}0\\2\\2\end{bmatrix}$ *and* $\begin{bmatrix}2\\2\\3\end{bmatrix}$

2. [Core]

Original: Chapter 1: Introduction to Vectors; Vectors and Linear Combinations; Problem 26.

母题覆盖：把向量组合转化为小型线性方程组，后面所有求系数题都用这个模板。

What combination $c\begin{bmatrix}2\\1\end{bmatrix}+d\begin{bmatrix}3\\1\end{bmatrix}$ produces $\begin{bmatrix}14\\8\end{bmatrix}$? Express this question as two equations for the coefficients c and d in the linear combination.

3. [Challenge]

Original: Chapter 1: Introduction to Vectors; Vectors and Linear Combinations; Problem 30.

母题覆盖：生成空间与线性相关的抽象母题，连接二维、四维和基的概念。

The linear combinations of $\boldsymbol{v} = (a, b)$ and $\boldsymbol{w} = (c, d)$ fill the plane unless __________. Find four vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}, \boldsymbol{z}$ with four components each so that their combinations $c\boldsymbol{u} + d\boldsymbol{v} + e\boldsymbol{w} + f\boldsymbol{z}$ produce all vectors (b_1, b_2, b_3, b_4) in four-dimensional space.

4. [Core]

Original: Chapter 1: Introduction to Vectors; Lengths and Dot Products; Problem 1.

母题覆盖：点积计算和点积线性性，长度、角度、投影都会反复用到。

Calculate the dot products $\boldsymbol{u} \cdot \boldsymbol{v}$ and $\boldsymbol{u} \cdot \boldsymbol{w}$ and $\boldsymbol{u} \cdot (\boldsymbol{v} + \boldsymbol{w})$ and $\boldsymbol{w} \cdot \boldsymbol{v}$:

$$\boldsymbol{u} = \begin{bmatrix}-.6\\.8\end{bmatrix}, \qquad \boldsymbol{v} = \begin{bmatrix}4\\3\end{bmatrix}, \qquad \boldsymbol{w} = \begin{bmatrix}1\\2\end{bmatrix}.$$

5. [Core]

Original: Chapter 1: Introduction to Vectors; Lengths and Dot Products; Problem 7.

母题覆盖：由点积求夹角，是所有角度判断题的标准流程。

Find the angle θ (from its cosine) between these pairs of vectors:

(a) $\boldsymbol{v} = \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ (b) $\boldsymbol{v} = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$

(c) $\boldsymbol{v} = \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix} -1 \\ \sqrt{3} \end{bmatrix}$ (d) $\boldsymbol{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$

6. [Core]

Original: Chapter 1: Introduction to Vectors; Lengths and Dot Products; Problem 6.

母题覆盖：垂直条件对应直线、平面、交线，后面四个基本子空间会继续使用。

(a) Describe every vector $\boldsymbol{w} = (w_1, w_2)$ that is perpendicular to $\boldsymbol{v} = (2, -1)$.

(b) All vectors perpendicular to $\boldsymbol{V} = (1, 1, 1)$ lie on a __________ in 3 dimensions.

(c) The vectors perpendicular to both $(1, 1, 1)$ and $(1, 2, 3)$ lie on a __________.

7. [Drill]

Original: Chapter 1: Introduction to Vectors; Lengths and Dot Products; Problem 12.

母题覆盖：投影系数的原型题，也是 Gram-Schmidt 的第一步。

With $\boldsymbol{v} = (1, 1)$ and $\boldsymbol{w} = (1, 5)$ choose a number c so that $\boldsymbol{w} - c\boldsymbol{v}$ is perpendicular to $\boldsymbol{v}$. Then find the formula for c starting from any nonzero $\boldsymbol{v}$ and $\boldsymbol{w}$.

8. [Core]

Original: Chapter 1: Introduction to Vectors; Matrices; Problem 1.

母题覆盖：矩阵乘法的列组合视角，后面列空间和线性方程组都靠它。

Find the linear combination $3\boldsymbol{s}_1 + 4\boldsymbol{s}_2 + 5\boldsymbol{s}_3 = \boldsymbol{b}$. Then write $\boldsymbol{b}$ as a matrix-vector multiplication $S\boldsymbol{x}$, with $3, 4, 5$ in $\boldsymbol{x}$. Compute the three dot products (row of S) $\cdot\boldsymbol{x}$:

$$\boldsymbol{s}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \boldsymbol{s}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \boldsymbol{s}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad \text{go into the columns of } S.$$

9. [Core]

Original: Chapter 1: Introduction to Vectors; Matrices; Problem 2.

母题覆盖：矩阵方程求系数，连接线性组合、方程组和逆矩阵。

Solve these equations $S\boldsymbol{y} = \boldsymbol{b}$ with $\boldsymbol{s}_1, \boldsymbol{s}_2, \boldsymbol{s}_3$ in the columns of S:

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 9 \end{bmatrix}.$$

S is a sum matrix. The sum of the first 3 odd numbers is __________.

10. [Drill]

Original: Chapter 1: Introduction to Vectors; Matrices; Problem 7.

母题覆盖：从 $A\boldsymbol{x} = 0$ 看列相关与行方程，是零空间章节的预备母题。

If the columns combine into $A\boldsymbol{x} = \boldsymbol{0}$, then each of the rows has $\boldsymbol{r} \cdot \boldsymbol{x} = 0$:

$$\begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{By rows} \quad \begin{bmatrix} \boldsymbol{r}_1 \cdot \boldsymbol{x} \\ \boldsymbol{r}_2 \cdot \boldsymbol{x} \\ \boldsymbol{r}_3 \cdot \boldsymbol{x} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The three rows also lie in a plane. Why is that plane perpendicular to $\boldsymbol{x}$?

2 Chapter 2: Solving Linear Equations

1. [Core]

Original: Chapter 2: Solving Linear Equations; Vectors and Linear Equations; Problem 9.

母题覆盖：行图像：用行点积计算 $A\boldsymbol{x}$，对应方程组每一行。

Compute each $A\boldsymbol{x}$ by dot products of the rows with the column vector:

$$\text{(a)} \begin{bmatrix} 1 & 2 & 4 \\ -2 & 3 & 1 \\ -4 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} \qquad \text{(b)} \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 2 \end{bmatrix}.$$

2. [Core]

Original: Chapter 2: Solving Linear Equations; Vectors and Linear Equations; Problem 10.

母题覆盖：列图像：同一个 $A\boldsymbol{x}$ 改成列向量组合，和上一题成对掌握。

Compute each $A\boldsymbol{x}$ in Problem 9 as a combination of the columns:

$$\text{9(a) becomes} \quad A\boldsymbol{x} = 2 \begin{bmatrix} 1 \\ -2 \\ -4 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} \\ \\ \\ \end{bmatrix}.$$

How many separate multiplications for $A\boldsymbol{x}$, when the matrix is "3 by 3"?

3. [Drill]

Original: Chapter 2: Solving Linear Equations; Vectors and Linear Equations; Problem 17.

母题覆盖：置换矩阵与基本线性变换，后面转置、置换和行交换都用这个想法。

Find the matrix P that multiplies (x, y, z) to give (y, x, z). Find the matrix Q that multiplies (y, x, z) to bring back (x, y, z).

4. [Core]

Original: Chapter 2: Solving Linear Equations; The Idea of Elimination; Problem 4.

母题覆盖：一般 2×2 消元公式：乘数、第二主元、奇异条件一次到位。

What multiple ℓ of equation 1 should be subtracted from equation 2 to remove c?

$$\begin{aligned} ax + by &= f \\ cx + dy &= g \end{aligned}$$

The first pivot is a (assumed nonzero). Elimination produces what formula for the second pivot? What is y? The second pivot is missing when $ad = bc$: singular.

5. [Core]

Original: Chapter 2: Solving Linear Equations; The Idea of Elimination; Problem 12.

母题覆盖：完整三元消元和回代，是所有手算消元题的基本流程。

Reduce this system to upper triangular form by two row operations:

$$\begin{aligned} 2x + 3y + z &= 8 \\ 4x + 7y + 5z &= 20 \\ -2y + 2z &= 0 \end{aligned}$$

Circle the pivots. Solve by back substitution for z, y, x.

6. [Core]

Original: Chapter 2: Solving Linear Equations; The Idea of Elimination; Problem 5.

母题覆盖：奇异系统的无解与无穷多解，训练对右端项的条件判断。

Choose a right side which gives no solution and another right side which gives infinitely many solutions. What are two of those solutions?

Singular system

$$\begin{aligned} 3x + 2y &= 10 \\ 6x + 4y &= \end{aligned}$$

7. [Core]

Original: Chapter 2: Solving Linear Equations; Elimination Using Matrices; Problem 1.

母题覆盖：把行操作写成消元矩阵，后面 $EA = U$、LU 都从这里来。

Write down the 3 by 3 matrices that produce these elimination steps:

(a) E_{21} subtracts 5 times row 1 from row 2.

(b) E_{32} subtracts -7 times row 2 from row 3.

(c) P exchanges rows 1 and 2, then rows 2 and 3.

8. [Drill]

Original: Chapter 2: Solving Linear Equations; Elimination Using Matrices; Problem 24.

母题覆盖：增广矩阵母题：判断可解、三角化、读出解。

Apply elimination to the 2 by 3 augmented matrix $[\,A \mid \boldsymbol{b}\,]$. What is the triangular system $U\boldsymbol{x} = \boldsymbol{c}$? What is the solution $\boldsymbol{x}$?

$$A\boldsymbol{x} = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 17 \end{bmatrix}.$$

9. [Core]

Original: Chapter 2: Solving Linear Equations; Rules for Matrix Operations; Problem 7.

母题覆盖：矩阵运算规则与反例，覆盖常见真假判断题。

True or false. Give a specific example when false:

(a) If columns 1 and 3 of B are the same, so are columns 1 and 3 of AB.

(b) If rows 1 and 3 of B are the same, so are rows 1 and 3 of AB.

(c) If rows 1 and 3 of A are the same, so are rows 1 and 3 of ABC.

(d) $(AB)^2 = A^2B^2$.

10. [Core]

Original: Chapter 2: Solving Linear Equations; Inverse Matrices; Problem 1.

母题覆盖：小矩阵求逆，建立逆矩阵的直接计算手感。

Find the inverses (directly or from the 2 by 2 formula) of A, B, C:

$$A = \begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & 0 \\ 4 & 2 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 3 & 4 \\ 5 & 7 \end{bmatrix}.$$

11. [Core]

Original: Chapter 2: Solving Linear Equations; Inverse Matrices; Problem 22.

母题覆盖：Gauss-Jordan 求逆母题，比公式更通用。

Change I into A^{-1} as you reduce A to I (by row operations):

$$[A \mid I] = \begin{bmatrix} 1 & 3 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{bmatrix} \quad \text{and} \quad [A \mid I] = \begin{bmatrix} 1 & 4 & 1 & 0 \\ 3 & 9 & 0 & 1 \end{bmatrix}.$$

12. [Core]

Original: Chapter 2: Solving Linear Equations; Elimination = Factorization: $A = LU$; Problem 5.

母题覆盖：从消元矩阵反推 $A = LU$，是 LU 分解的核心模板。

What matrix E puts A into triangular form $EA = U$? Multiply by $E^{-1} = L$ to factor A into LU:

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 2 \\ 6 & 3 & 5 \end{bmatrix}.$$

13. [Drill]

Original: Chapter 2: Solving Linear Equations; Elimination = Factorization: $A = LU$; Problem 16.

母题覆盖：用 LU 解方程：先前代再回代，覆盖分解后的求解题。

Solve $L\boldsymbol{c} = \boldsymbol{b}$. Then solve $U\boldsymbol{x} = \boldsymbol{c}$. What was A?

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{b} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}.$$

14. [Core]

Original: Chapter 2: Solving Linear Equations; Transposes and Permutations; Problem 2.

母题覆盖：转置乘法规则，覆盖 $(AB)^T$、顺序反转、对称性判断。

Verify that $(AB)^{\mathrm{T}}$ equals $B^{\mathrm{T}}A^{\mathrm{T}}$ but those are different from $A^{\mathrm{T}}B^{\mathrm{T}}$:

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}, \quad AB = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}.$$

Show also that AA^{T} is different from $A^{\mathrm{T}}A$. But both of those matrices are ________.

15. [Drill]

Original: Chapter 2: Solving Linear Equations; Transposes and Permutations; Problem 20.

母题覆盖：对称矩阵的 LDL^T 分解，连接消元、转置和正定矩阵。

Factor these symmetric matrices into $S = LDL^{\mathrm{T}}$. The pivot matrix D is diagonal:

$$S = \begin{bmatrix} 1 & 3 \\ 3 & 2 \end{bmatrix}, \quad S = \begin{bmatrix} 1 & b \\ b & c \end{bmatrix}, \quad S = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}.$$

3 Chapter 3: Vector Spaces and Subspaces

1. [Core]

Original: Chapter 3: Vector Spaces and Subspaces; Spaces of Vectors; Problem 18.

母题覆盖：子空间三项检查：零向量、加法封闭、数乘封闭。

Which of the following subsets of $\mathbb{R}^3$ are actually subspaces?

(a) The plane of vectors (b_1, b_2, b_3) with $b_1 = b_2$.

(b) The plane of vectors with $b_1 = 1$.

(c) The vectors with $b_1 b_2 b_3 = 0$.

(d) All linear combinations of $v = (1, 4, 0)$ and $w = (2, 2, 2)$.

(e) All vectors that satisfy $b_1 + b_2 + b_3 = 0$.

(f) All vectors with $b_1 \le b_2 \le b_3$.

2. [Core]

Original: Chapter 3: Vector Spaces and Subspaces; Spaces of Vectors; Problem 28.

母题覆盖：列空间与可解条件，训练“哪些右端项可达”。

For which right sides (find a condition on b_1, b_2, b_3) are these systems solvable?

(a) $$\begin{bmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ -1 & -4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

(b) $$\begin{bmatrix} 1 & 4 \\ 2 & 9 \\ -1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

3. [Core]

Original: Chapter 3: Vector Spaces and Subspaces; The Nullspace of A : Solving $Ax = 0$ and $Rx = 0$; Problem 1.

母题覆盖：从矩阵到阶梯形，识别主元变量和自由变量。

Reduce A and B to their triangular echelon forms U. Which variables are free?

(a) $A = \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 1 & 2 & 3 & 6 & 9 \\ 0 & 0 & 1 & 2 & 3 \end{bmatrix}$

(b) $B = \begin{bmatrix} 2 & 4 & 2 \\ 0 & 4 & 4 \\ 0 & 8 & 8 \end{bmatrix}$

4. [Core]

Original: Chapter 3: Vector Spaces and Subspaces; The Nullspace of A : Solving $Ax = 0$ and $Rx = 0$; Problem 2.

母题覆盖：特殊解母题：每个自由变量产生一个零空间基向量。

For the matrices in Problem 1, find a special solution for each free variable. (Set the free variable to 1. Set the other free variables to zero.)

5. [Drill]

Original: Chapter 3: Vector Spaces and Subspaces; The Nullspace of A : Solving $Ax = 0$ and $Rx = 0$; Problem 10.

母题覆盖：计数定理 $r + (n - r) = n$，覆盖主元数、自由变量数、零空间维数。

Suppose an m by n matrix has r pivots. The number of special solutions is __________. The nullspace contains only $x = 0$ when $r =$ __________. The column space is all of $\mathbb{R}^m$ when $r =$ __________.

6. [Core]

Original: Chapter 3: Vector Spaces and Subspaces; The Complete Solution to $Ax = b$; Problem 1.

母题覆盖：完整解母题：列空间、零空间、特解、通解一次完成。

(Recommended) Execute the six steps of Worked Example 3.3 A to describe the column space and nullspace of A and the complete solution to $A\mathbf{x} = \mathbf{b}$:

$$A = \begin{bmatrix} 2 & 4 & 6 & 4 \\ 2 & 5 & 7 & 6 \\ 2 & 3 & 5 & 2 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}.$$

7. [Drill]

Original: Chapter 3: Vector Spaces and Subspaces; The Complete Solution to $Ax = b$; Problem 2.

母题覆盖：秩一矩阵的可解条件，是列空间条件最清楚的特例。

Carry out the same six steps for this matrix A with rank one. You will find *two* conditions on b_1, b_2, b_3 for $A\mathbf{x} = \mathbf{b}$ to be solvable. Together these two conditions put $\mathbf{b}$ into the __________ space (two planes give a line):

$$A = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 3 \\ 6 & 3 & 9 \\ 4 & 2 & 6 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 30 \\ 20 \end{bmatrix}.$$

8. [Drill]

Original: Chapter 3: Vector Spaces and Subspaces; The Complete Solution to $Ax = b$; Problem 5.

母题覆盖：带参数右端项的可解条件，训练增广矩阵判别。

Under what condition on b_1, b_2, b_3 is this system solvable? Include $\mathbf{b}$ as a fourth column in elimination. Find all solutions when that condition holds:

$$\begin{aligned} x + 2y - 2z &= b_1 \\ 2x + 5y - 4z &= b_2 \\ 4x + 9y - 8z &= b_3. \end{aligned}$$

9. [Core]

Original: Chapter 3: Vector Spaces and Subspaces; Independence, Basis and Dimension; Problem 5.

母题覆盖：线性相关与无关判断，覆盖向量组是否为基的前置步骤。

Decide the dependence or independence of

(a) the vectors $(1, 3, 2)$ and $(2, 1, 3)$ and $(3, 2, 1)$

(b) the vectors $(1, -3, 2)$ and $(2, 1, -3)$ and $(-3, 2, 1)$.

10. [Core]

Original: Chapter 3: Vector Spaces and Subspaces; Independence, Basis and Dimension; Problem 16.

母题覆盖：给子空间找基和维数，覆盖平面、直线、方程约束型子空间。

Find a basis for each of these subspaces of $\mathbf{R}^4$:

(a) All vectors whose components are equal.

(b) All vectors whose components add to zero.

(c) All vectors that are perpendicular to $(1,1,0,0)$ and $(1,0,1,1)$.

(d) The column space and the nullspace of I (4 by 4).

11. [Core]

Original: Chapter 3: Vector Spaces and Subspaces; Dimensions of the Four Subspaces; Problem 2.

母题覆盖：四个基本子空间的基与维数，是第三章最核心的综合母题。

Find bases and dimensions for the four subspaces associated with A and B:

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 5 & 8 \end{bmatrix}.$$

12. [Challenge]

Original: Chapter 3: Vector Spaces and Subspaces; Dimensions of the Four Subspaces; Problem 11.

母题覆盖：一般 $m \times n$、秩 r 情形下的可解性与左零空间，是 Fredholm 思想的入口。

(Important) A is an m by n matrix of rank r. Suppose there are right sides $\boldsymbol{b}$ for which $A\boldsymbol{x} = \boldsymbol{b}$ has *no solution*.

(a) What are all inequalities ($<$ or $\leq$) that must be true between m, n, and r?

(b) How do you know that $A^{\mathrm{T}}\boldsymbol{y} = \mathbf{0}$ has solutions other than $\boldsymbol{y} = \mathbf{0}$?

4 Chapter 4: Orthogonality

1. [Core]

Original: Chapter 4: Orthogonality; Orthogonality of the Four Subspaces; Problem 6.

母题覆盖：无解系统中的左零空间证据，覆盖“不相容”的标准证明。

This system of equations $A\boldsymbol{x} = \boldsymbol{b}$ has *no solution* (they lead to $0 = 1$):

$$\begin{aligned} x + 2y + 2z &= 5 \\ 2x + 2y + 3z &= 5 \\ 3x + 4y + 5z &= 9 \end{aligned}$$

Find numbers y_1, y_2, y_3 to multiply the equations so they add to $0 = 1$. You have found a vector $\boldsymbol{y}$ in which subspace? Its dot product $\boldsymbol{y}^{\mathrm{T}}\boldsymbol{b}$ is 1, so no solution $\boldsymbol{x}$.

2. [Core]

Original: Chapter 4: Orthogonality; Orthogonality of the Four Subspaces; Problem 16.

母题覆盖：证明 $N(A^T)$ 垂直于 $C(A)$，四个基本子空间正交关系的母题。

Prove that every $\boldsymbol{y}$ in $\mathbf{N}(A^{\mathrm{T}})$ is perpendicular to every $A\boldsymbol{x}$ in the column space, using the matrix shorthand of equation (2). Start from $A^{\mathrm{T}}\boldsymbol{y} = \mathbf{0}$.

3. [Core]

Original: Chapter 4: Orthogonality; Projections; Problem 1.

母题覆盖：投影到直线：求 $\hat{x}$、p、误差 e，所有投影题的最小模板。

Project the vector $\boldsymbol{b}$ onto the line through $\boldsymbol{a}$. Check that $\boldsymbol{e}$ is perpendicular to $\boldsymbol{a}$:

$$\text{(a)} \quad \boldsymbol{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \quad \text{and} \quad \boldsymbol{a} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \qquad \text{(b)} \quad \boldsymbol{b} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} \quad \text{and} \quad \boldsymbol{a} = \begin{bmatrix} -1 \\ -3 \\ -1 \end{bmatrix}.$$

4. [Drill]

Original: Chapter 4: Orthogonality; Projections; Problem 3.

母题覆盖：投影矩阵 $P = aa^T/(a^Ta)$，覆盖 $P^2 = P$ 和 $Pb = p$。

In Problem 1, find the projection matrix $P = \boldsymbol{a}\boldsymbol{a}^{\mathrm{T}}/\boldsymbol{a}^{\mathrm{T}}\boldsymbol{a}$ onto the line through each vector $\boldsymbol{a}$. Verify in both cases that $P^2 = P$. Multiply $P\boldsymbol{b}$ in each case to compute the projection $\boldsymbol{p}$.

5. [Core]

Original: Chapter 4: Orthogonality; Projections; Problem 11.

母题覆盖：投影到列空间：解正规方程，连接投影和最小二乘。

Project $\boldsymbol{b}$ onto the column space of A by solving $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$ and $\boldsymbol{p} = A\widehat{\boldsymbol{x}}$:

$$\text{(a)} \quad A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad \boldsymbol{b} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \qquad \text{(b)} \quad A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad \boldsymbol{b} = \begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix}.$$

Find $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$. It should be perpendicular to the columns of A.

6. [Core]

Original: Chapter 4: Orthogonality; Least Squares Approximations; Problem 1.

母题覆盖：最小二乘拟合直线的完整流程：构造 A、正规方程、投影和误差。

With $\boldsymbol{b} = (0, 8, 8, 20)$ at $t = 0, 1, 3, 4$, set up and solve the normal equations $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$. For the best straight line in Figure 4.9a, find its four heights p_i and four errors e_i. What is the minimum value $E = e_1^2 + e_2^2 + e_3^2 + e_4^2$?

7. [Drill]

Original: Chapter 4: Orthogonality; Least Squares Approximations; Problem 4.

母题覆盖：从误差平方和求导推出正规方程，覆盖计算型和推导型最小二乘题。

(By calculus) Write down $E = \|A\boldsymbol{x} - \boldsymbol{b}\|^2$ as a sum of four squares—the last one is $(C + 4D - 20)^2$. Find the derivative equations $\partial E/\partial C = 0$ and $\partial E/\partial D = 0$. Divide by 2 to obtain the normal equations $A^{\mathrm{T}}A\widehat{\boldsymbol{x}} = A^{\mathrm{T}}\boldsymbol{b}$.

8. [Drill]

Original: Chapter 4: Orthogonality; Least Squares Approximations; Problem 21.

母题覆盖：误差向量属于哪个子空间，巩固最小二乘的几何解释。

Which of the four subspaces contains the error vector $\boldsymbol{e}$? Which contains $\boldsymbol{p}$? Which contains $\widehat{\boldsymbol{x}}$? What is the nullspace of A?

9. [Core]

Original: Chapter 4: Orthogonality; Orthonormal Bases and Gram-Schmidt; Problem 13.

母题覆盖：Gram-Schmidt 第一步：减去投影得到正交向量。

What multiple of $\boldsymbol{a} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ should be subtracted from $\boldsymbol{b} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$ to make the result $\boldsymbol{B}$ orthogonal to $\boldsymbol{a}$? Sketch a figure to show $\boldsymbol{a}, \boldsymbol{b}$, and $\boldsymbol{B}$.

10. [Core]

Original: Chapter 4: Orthogonality; Orthonormal Bases and Gram-Schmidt; Problem 14.

母题覆盖：完成 Gram-Schmidt 并写出 QR，覆盖正交化到分解的完整链条。

Complete the Gram-Schmidt process in Problem 13 by computing $\boldsymbol{q}_1 = \boldsymbol{a}/\|\boldsymbol{a}\|$ and $\boldsymbol{q}_2 = \boldsymbol{B}/\|\boldsymbol{B}\|$ and factoring into QR:

$$\begin{bmatrix} 1 & 4 \\ 1 & 0 \end{bmatrix} = [\,\boldsymbol{q}_1 \quad \boldsymbol{q}_2\,] \begin{bmatrix} \|\boldsymbol{a}\| & ? \\ 0 & \|\boldsymbol{B}\| \end{bmatrix}.$$

11. [Drill]

Original: Chapter 4: Orthogonality; Orthonormal Bases and Gram-Schmidt; Problem 7.

母题覆盖：当 Q 有正交规范列时，最小二乘直接化为 $Q^T b$。

If Q has orthonormal columns, what is the least squares solution $\widehat{\boldsymbol{x}}$ to $Q\boldsymbol{x} = \boldsymbol{b}$?

5 Chapter 5: Determinants

1. [Core]

Original: Chapter 5: Determinants; The Properties of Determinants; Problem 13.

母题覆盖：用消元和主元求行列式，覆盖大多数计算题。

Reduce A to U and find $\det A$ = product of the pivots:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix} \qquad A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & 3 \\ 3 & 3 & 3 \end{bmatrix}.$$

2. [Core]

Original: Chapter 5: Determinants; The Properties of Determinants; Problem 28.

母题覆盖：行列式性质真假判断，覆盖乘法、转置、逆、行操作等常见变形。

True or false (give a reason if true or a 2 by 2 example if false):

(a) If A is not invertible then AB is not invertible.

(b) The determinant of A is always the product of its pivots.

(c) The determinant of $A - B$ equals $\det A - \det B$.

(d) AB and BA have the same determinant.

3. [Core]

Original: Chapter 5: Determinants; Permutations and Cofactors; Problem 1.

母题覆盖：按排列的六项公式计算 3×3 行列式，理解符号来源。

Compute the determinants of A, B, C from six terms. Are their rows independent?

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 3 & 2 & 1 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 4 & 4 \\ 5 & 6 & 7 \end{bmatrix} \qquad C = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

4. [Drill]

Original: Chapter 5: Determinants; Permutations and Cofactors; Problem 11.

母题覆盖：cofactor matrix 与伴随矩阵，覆盖按余子式求逆的前置操作。

Find all cofactors and put them into cofactor matrices C, D. Find AC and $\det B$.

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 0 & 0 \end{bmatrix}.$$

5. [Core]

Original: Chapter 5: Determinants; Cramer's Rule, Inverses, and Volumes; Problem 1.

母题覆盖：Cramer 法则母题，覆盖用行列式解线性方程组。

Solve these linear equations by Cramer's Rule $x_j = \det B_j / \det A$:

$$\text{(a)} \quad \begin{aligned} 2x_1 + 5x_2 &= 1 \\ x_1 + 4x_2 &= 2 \end{aligned} \qquad\qquad \text{(b)} \quad \begin{aligned} 2x_1 + x_2 &= 1 \\ x_1 + 2x_2 + x_3 &= 0 \\ x_2 + 2x_3 &= 0 \end{aligned}$$

6. [Core]

Original: Chapter 5: Determinants; Cramer's Rule, Inverses, and Volumes; Problem 6.

母题覆盖：用 cofactor formula 求逆，连接余子式、伴随矩阵和逆矩阵。

Find A^{-1} from the cofactor formula $C^{\mathrm{T}} / \det A$. Use symmetry in part (b).

$$\text{(a)} \quad A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 0 \\ 0 & 7 & 1 \end{bmatrix} \qquad \text{(b)} \quad A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}.$$

7. [Drill]

Original: Chapter 5: Determinants; Cramer's Rule, Inverses, and Volumes; Problem 17.

母题覆盖：行列式的体积解释，覆盖面积、体积、几何缩放类题。

A box has edges from $(0,0,0)$ to $(3,1,1)$ and $(1,3,1)$ and $(1,1,3)$. Find its volume. Also find the area of each parallelogram face using $\|\boldsymbol{u} \times \boldsymbol{v}\|$.

6 Chapter 6: Eigenvalues and Eigenvectors

1. [Core]

Original: Chapter 6: Eigenvalues and Eigenvectors; Introduction to Eigenvalues; Problem 2.

母题覆盖：求特征值和特征向量的标准流程。

Find the eigenvalues and the eigenvectors of these two matrices:

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \quad \text{and} \quad A + I = \begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}.$$

$A + I$ has the _____ eigenvectors as A. Its eigenvalues are _____ by 1.

2. [Core]

Original: Chapter 6: Eigenvalues and Eigenvectors; Introduction to Eigenvalues; Problem 10.

母题覆盖：Markov 矩阵和稳定状态，覆盖矩阵幂长期行为。

Find the eigenvalues and eigenvectors for both of these Markov matrices A and A^∞. Explain from those answers why A^{100} is close to A^∞:

$$A = \begin{bmatrix} .6 & .2 \\ .4 & .8 \end{bmatrix} \quad \text{and} \quad A^\infty = \begin{bmatrix} 1/3 & 1/3 \\ 2/3 & 2/3 \end{bmatrix}.$$

3. [Drill]

Original: Chapter 6: Eigenvalues and Eigenvectors; Introduction to Eigenvalues; Problem 16.

母题覆盖：行列式等于特征值乘积，连接特征值和行列式。

The determinant of A equals the product $\lambda_1\lambda_2\cdots\lambda_n$. Start with the polynomial $\det(A - \lambda I)$ separated into its n factors (always possible). Then set $\lambda = 0$:

$$\det(A - \lambda I) = (\lambda_1 - \lambda)(\lambda_2 - \lambda)\cdots(\lambda_n - \lambda) \quad \text{so} \quad \det A = \_\_\_\_\_.$$

Check this rule in Example 1 where the Markov matrix has $\lambda = 1$ and $\frac{1}{2}$.

4. [Drill]

Original: Chapter 6: Eigenvalues and Eigenvectors; Introduction to Eigenvalues; Problem 17.

母题覆盖：迹等于特征值之和，覆盖快速检查和参数题。

The sum of the diagonal entries (the *trace*) equals the sum of the eigenvalues:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{has} \quad \det(A - \lambda I) = \lambda^2 - (a+d)\lambda + ad - bc = 0.$$

The quadratic formula gives the eigenvalues $\lambda = (a+d+\sqrt{\quad})/2$ and $\lambda =$ _____. Their sum is _____. If A has $\lambda_1 = 3$ and $\lambda_2 = 4$ then $\det(A - \lambda I) =$ _____.

5. [Core]

Original: Chapter 6: Eigenvalues and Eigenvectors; Diagonalizing a Matrix; Problem 8.

母题覆盖：对角化并计算矩阵幂，Fibonacci 是典型综合题。

Diagonalize the Fibonacci matrix by completing X^{-1}:

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix}.$$

Do the multiplication $X\Lambda^k X^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ to find its second component. This is the kth Fibonacci number $F_k = (\lambda_1^k - \lambda_2^k)/(\lambda_1 - \lambda_2)$.

6. [Core]

Original: Chapter 6: Eigenvalues and Eigenvectors; Diagonalizing a Matrix; Problem 14.

母题覆盖：重复特征值但不可对角化，是判断对角化失败的母题。

The matrix $A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ is not diagonalizable because the rank of $A - 3I$ is _____ . Change one entry to make A diagonalizable. Which entries could you change?

7. [Drill]

Original: Chapter 6: Eigenvalues and Eigenvectors; Diagonalizing a Matrix; Problem 20.

母题覆盖：从 $A = X\Lambda X^{-1}$ 推出行列式性质，训练相似变换思维。

Suppose $A = X\Lambda X^{-1}$. Take determinants to prove $\det A = \det \Lambda = \lambda_1\lambda_2\cdots\lambda_n$. This quick proof only works when A can be _____ .

8. [Core]

Original: Chapter 6: Eigenvalues and Eigenvectors; Systems of Differential Equations; Problem 1.

母题覆盖：用 $e^{\lambda t}x$ 解线性微分方程组。

Find two λ's and x's so that $u = e^{\lambda t}x$ solves

$$\frac{du}{dt} = \begin{bmatrix} 4 & 3 \\ 0 & 1 \end{bmatrix} u.$$

What combination $u = c_1 e^{\lambda_1 t} x_1 + c_2 e^{\lambda_2 t} x_2$ starts from $u(0) = (5, -2)$?

9. [Core]

Original: Chapter 6: Eigenvalues and Eigenvectors; Systems of Differential Equations; Problem 12.

母题覆盖：重复根和第二解，覆盖不可对角化动态系统的入口。

Substitute $y = e^{\lambda t}$ into $y'' = 6y' - 9y$ to show that $\lambda = 3$ is a repeated root. This is trouble; we need a second solution after e^{3t}. The matrix equation is

$$\frac{d}{dt}\begin{bmatrix} y \\ y' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -9 & 6 \end{bmatrix} \begin{bmatrix} y \\ y' \end{bmatrix}.$$

Show that this matrix has $\lambda = 3, 3$ and only one line of eigenvectors. *Trouble here too.* Show that the second solution to $y'' = 6y' - 9y$ is $y = te^{3t}$.

10. [Challenge]

Original: Chapter 6: Eigenvalues and Eigenvectors; Systems of Differential Equations; Problem 26.

母题覆盖：矩阵指数永远可逆，连接 e^{At}、特征值和系统演化。

(Recommended) Give two reasons why the matrix exponential e^{At} is never singular:

(a) Write down its inverse.

(b) Why are these eigenvalues nonzero? If $Ax = \lambda x$ then $e^{At}x = \underline{\hspace{2em}}x$.

11. [Core]

Original: Chapter 6: Eigenvalues and Eigenvectors; Symmetric Matrices; Problem 6.

母题覆盖：对称矩阵的正交对角化，覆盖 $S = Q\Lambda Q^T$ 的完整算法。

Find an orthogonal matrix Q that diagonalizes $S = \begin{bmatrix} -2 & 6 \\ 6 & 7 \end{bmatrix}$. What is Λ?

12. [Drill]

Original: Chapter 6: Eigenvalues and Eigenvectors; Symmetric Matrices; Problem 20.

母题覆盖：对称矩阵不同特征值的特征向量正交，是核心证明题。

Another proof that eigenvectors are perpendicular when $S = S^{\mathrm{T}}$. Two steps:

(a) Suppose $Sx = \lambda x$ and $Sy = 0y$ and $\lambda \neq 0$. Then y is in the nullspace and x is in the column space. They are perpendicular because _____. Go carefully—why are these subspaces orthogonal?

(b) If $Sy = \beta y$, apply that argument to $S - \beta I$. One eigenvalue of $S - \beta I$ moves to zero. The eigenvectors x, y stay the same—so they are perpendicular.

13. [Challenge]

Original: Chapter 6: Eigenvalues and Eigenvectors; Symmetric Matrices; Problem 32.

母题覆盖：Rayleigh quotient 最大值，连接对称矩阵、优化和正定性。

If $\lambda_{\max}$ is the largest eigenvalue of a symmetric matrix S, no diagonal entry can be larger than $\lambda_{\max}$. What is the first entry a_{11} of $S = Q\Lambda Q^{\mathrm{T}}$? Show why $a_{11} \leq \lambda_{\max}$.

14. [Core]

Original: Chapter 6: Eigenvalues and Eigenvectors; Positive Definite Matrices; Problem 3.

母题覆盖：正定参数题，覆盖 2×2 判据、主元和 LDL^T。

For which numbers b and c are these matrices positive definite?

$$S = \begin{bmatrix} 1 & b \\ b & 9 \end{bmatrix} \quad S = \begin{bmatrix} 2 & 4 \\ 4 & c \end{bmatrix} \quad S = \begin{bmatrix} c & b \\ b & c \end{bmatrix}.$$

With the pivots in D and multiplier in L, factor each A into LDL^{T}.

15. [Core]

Original: Chapter 6: Eigenvalues and Eigenvectors; Positive Definite Matrices; Problem 11.

母题覆盖：三阶正定判据：左上角主子式、主元和行列式比值。

Compute the three upper left determinants of S to establish positive definiteness. Verify that their ratios give the second and third pivots.

$$\text{Pivots} = \text{ratios of determinants} \quad S = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 3 \\ 0 & 3 & 8 \end{bmatrix}.$$

16. [Drill]

Original: Chapter 6: Eigenvalues and Eigenvectors; Positive Definite Matrices; Problem 14.

母题覆盖：证明 S^{-1} 正定，覆盖正定性的变换证明。

If S is positive definite then S^{-1} is positive definite. Best proof: The eigenvalues of S^{-1} are positive because _____. Second proof (only for 2 by 2):

$$\text{The entries of } S^{-1} = \frac{1}{ac-b^2}\begin{bmatrix} c & -b \\ -b & a \end{bmatrix} \text{ pass the determinant tests } \_\_\_\_\_.$$

17. [Core]

Original: Chapter 6: Eigenvalues and Eigenvectors; Positive Definite Matrices; Problem 25.

母题覆盖：Cholesky 分解母题，连接正定矩阵和平方根分解。

With positive pivots in D, the factorization $S = LDL^{\mathrm{T}}$ becomes $L\sqrt{D}\sqrt{D}L^{\mathrm{T}}$. (Square roots of the pivots give $D = \sqrt{D}\sqrt{D}$.) Then $C = \sqrt{D}L^{\mathrm{T}}$ yields the **Cholesky factorization** $A = C^{\mathrm{T}}C$ which is "symmetrized LU":

$$\text{From } C = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \text{ find } S. \quad \text{From } S = \begin{bmatrix} 4 & 8 \\ 8 & 25 \end{bmatrix} \text{ find } C = \mathbf{chol}(S).$$

7 Chapter 7: The Singular Value Decomposition (SVD)

1. [Core]

Original: Chapter 7: The Singular Value Decomposition (SVD); Image Processing by Linear Algebra; Problem 1.

母题覆盖：秩和秩一分解，SVD 低秩近似的前置母题。

What are the ranks r for these matrices with entries i times j and i plus j? Write A and B as the sum of r pieces $\boldsymbol{u}\boldsymbol{v}^{\mathrm{T}}$ of rank one. Not requiring $\boldsymbol{u}_1^{\mathrm{T}}\boldsymbol{u}_2 = \boldsymbol{v}_1^{\mathrm{T}}\boldsymbol{v}_2 = \mathbf{0}$.

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \\ 4 & 8 & 12 & 16 \end{bmatrix} \qquad B = \begin{bmatrix} 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \\ 5 & 6 & 7 & 8 \end{bmatrix}$$

2. [Core]

Original: Chapter 7: The Singular Value Decomposition (SVD); Image Processing by Linear Algebra; Problem 6.

母题覆盖：区分 eigenvalues 与 singular values，覆盖 A^TA 的核心作用。

Find the eigenvalues and the singular values of this 2 by 2 matrix A.

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} \qquad \text{with} \quad A^{\mathrm{T}}A = \begin{bmatrix} 20 & 10 \\ 10 & 5 \end{bmatrix} \qquad \text{and} \quad AA^{\mathrm{T}} = \begin{bmatrix} 5 & 10 \\ 10 & 20 \end{bmatrix}.$$

The eigenvectors $(1, 2)$ and $(1, -2)$ of A are not orthogonal. How do you know the eigenvectors $\boldsymbol{v}_1, \boldsymbol{v}_2$ of $A^{\mathrm{T}}A$ are orthogonal? Notice that $A^{\mathrm{T}}A$ and AA^{T} have the same eigenvalues (25 and 0).

3. [Drill]

Original: Chapter 7: The Singular Value Decomposition (SVD); Image Processing by Linear Algebra; Problem 7.

母题覆盖：从 $A = U\Sigma V^T$ 推出 $AV = U\Sigma$，是计算 SVD 的操作形式。

How does the second form $A\boldsymbol{V} = \boldsymbol{U}\Sigma$ in equation (3) follow from the first form $A = \boldsymbol{U}\Sigma\boldsymbol{V}^{\mathrm{T}}$? That is the most famous form of the SVD.

4. [Core]

Original: Chapter 7: The Singular Value Decomposition (SVD); Bases and Matrices in the SVD; Problem 1.

母题覆盖：从 A^TA 和 AA^T 构造 SVD 的最小模板。

Find the eigenvalues of these matrices. Then find singular values from $A^{\mathrm{T}}A$:

$$A = \begin{bmatrix} 0 & 4 \\ 0 & 0 \end{bmatrix} \qquad A = \begin{bmatrix} 0 & 4 \\ 1 & 0 \end{bmatrix}$$

For each A, construct V from the eigenvectors of $A^{\mathrm{T}}A$ and U from the eigenvectors of AA^{T}. Check that $A = U\Sigma V^{\mathrm{T}}$.

5. [Core]

Original: Chapter 7: The Singular Value Decomposition (SVD); Bases and Matrices in the SVD; Problem 2.

母题覆盖：完整 2×2 SVD 计算题，覆盖 V, Σ, U 的顺序。

Find $A^{\mathrm{T}}A$ and V and Σ and $\boldsymbol{u}_i = A\boldsymbol{v}_i/\sigma_i$ and the full SVD:

$$A = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} = U\Sigma V^{\mathrm{T}}.$$

6. [Core]

Original: Chapter 7: The Singular Value Decomposition (SVD); Bases and Matrices in the SVD; Problem 4.

母题覆盖：矩形矩阵 SVD，覆盖 Σ 形状和非零奇异值。

Compute $A^{\mathrm{T}}A$ and AA^{T} and their eigenvalues and unit eigenvectors for V and U.

$$\textbf{Rectangular matrix} \qquad A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Check $AV = U\Sigma$ (this decides $\pm$ signs in U). Σ has the same shape as $A : 2 \times 3$.

7. [Drill]

Original: Chapter 7: The Singular Value Decomposition (SVD); Bases and Matrices in the SVD; Problem 6.

母题覆盖：证明 A^TA 与 AA^T 非零特征值相同，解释奇异值来源。

Substitute the SVD for A and A^{T} to show that $A^{\mathrm{T}}A$ has its eigenvalues in $\Sigma^{\mathrm{T}}\Sigma$ and AA^{T} has its eigenvalues in $\Sigma\Sigma^{\mathrm{T}}$. Since a diagonal $\Sigma^{\mathrm{T}}\Sigma$ has the same nonzeros as $\Sigma\Sigma^{\mathrm{T}}$, we see again that $A^{\mathrm{T}}A$ and AA^{T} have the same nonzero eigenvalues.

8. [Challenge]

Original: Chapter 7: The Singular Value Decomposition (SVD); Bases and Matrices in the SVD; Problem 19.

母题覆盖：最近奇异矩阵，覆盖最小奇异值和低秩扰动思想。

Suppose A is invertible (with $\sigma_1 > \sigma_2 > 0$). Change A by ***as small a matrix as possible*** to produce a singular matrix A_0. Hint: U and V do not change:

$$\text{From} \qquad A = \begin{bmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 \end{bmatrix} \begin{bmatrix} \sigma_1 & \\ & \sigma_2 \end{bmatrix} \begin{bmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 \end{bmatrix}^{\mathrm{T}} \qquad \text{find the nearest } A_0.$$

9. [Challenge]

Original: Chapter 7: The Singular Value Decomposition (SVD); Bases and Matrices in the SVD; Problem 26.

母题覆盖：SVD 的几何解释：圆到椭圆，覆盖旋转、拉伸、再旋转。

Every matrix $A = U\Sigma V^{\mathrm{T}}$ takes **circles to ellipses**. $A\boldsymbol{V} = U\Sigma$ says that the radius vectors $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ of the circle go to the semi-axes $\sigma_1\boldsymbol{u}_1$ and $\sigma_2\boldsymbol{u}_2$ of the ellipse. Draw the circle and the ellipse for $\theta = 30°$:

$$V = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \qquad U = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \qquad \Sigma = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}.$$

Section 7.4 will start with an important SVD picture for 2 by 2 matrices:

$$A = (\textbf{rotate})(\textbf{stretch})(\textbf{rotate}).$$

With symmetry

$$S = (\textbf{rotate})(\textbf{stretch})(\textbf{rotate back}).$$

10. [Core]

Original: Chapter 7: The Singular Value Decomposition (SVD); Principal Component Analysis (PCA by the SVD); Problem 1.

母题覆盖：PCA 完整流程：中心化、协方差、主成分方向。

Suppose A_0 holds these 2 measurements of 5 samples:

$$A_0 = \begin{bmatrix} 5 & 4 & 3 & 2 & 1 \\ -1 & 1 & 0 & 1 & -1 \end{bmatrix}$$

Find the average of each row and subtract it to produce the centered matrix A. Compute the sample covariance matrix $S = AA^{\mathrm{T}}/(n-1)$ and find its eigenvalues λ_1 and λ_2. What line through the origin is closest to the 5 samples in columns of A?

11. [Drill]

Original: Chapter 7: The Singular Value Decomposition (SVD); Principal Component Analysis (PCA by the SVD); Problem 5.

母题覆盖：从协方差到相关矩阵，覆盖标准化变量的 PCA 前处理。

From this sample covariance matrix S, find the correlation matrix DSD with 1's down its main diagonal. D is a positive diagonal matrix that produces those 1's.

$$S = \begin{bmatrix} 4 & 2 & 0 \\ 2 & 4 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

12. [Core]

Original: Chapter 7: The Singular Value Decomposition (SVD); The Geometry of the SVD; Problem 1.

母题覆盖：秩一矩阵完整 SVD，连接零奇异值和四个子空间。

(a) Compute $A^{\mathrm{T}}A$ and its eigenvalues and unit eigenvectors $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$. Find σ_1.

$$\textbf{Rank one matrix} \quad A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$$

(b) Compute AA^{T} and its eigenvalues and unit eigenvectors $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$.

(c) Verify that $A\boldsymbol{v}_1 = \sigma_1\boldsymbol{u}_1$. Put numbers into $\boldsymbol{A} = \boldsymbol{U\Sigma V}^{\mathbf{T}}$ (this is the SVD).

13. [Core]

Original: Chapter 7: The Singular Value Decomposition (SVD); The Geometry of the SVD; Problem 4.

母题覆盖：伪逆、投影 A^+A 与 AA^+，覆盖欠定/秩亏问题。

Compute the pseudoinverse $A^+ = V\Sigma^+U^\mathrm{T}$. The diagonal matrix Σ^+ contains $1/\sigma_1$. Rename the four subspaces (for A) in Figure 7.6 as four subspaces for A^+. Compute the projections A^+A and AA^+ on the row and column spaces of A.

14. [Challenge]

Original: Chapter 7: The Singular Value Decomposition (SVD); The Geometry of the SVD; Problem 16.

母题覆盖：最短最小二乘解，是伪逆应用的终点母题。

The vector $\boldsymbol{x}^+ = A^+\boldsymbol{b}$ *is the shortest possible solution to* $A^\mathrm{T}A\widehat{\boldsymbol{x}} = A^\mathrm{T}\boldsymbol{b}$.
Reason: The difference $\widehat{\boldsymbol{x}} - \boldsymbol{x}^+$ is in the nullspace of $A^\mathrm{T}A$. This is also the nullspace of A, orthogonal to $\boldsymbol{x}^+$. Explain how it follows that $\|\widehat{\boldsymbol{x}}\|^2 = \|\boldsymbol{x}^+\|^2 + \|\widehat{\boldsymbol{x}} - \boldsymbol{x}^+\|^2$.

复习建议

- 第一轮只做 [Core]，每题都总结它的“可复用步骤”。
- 第二轮补 [Drill]，检查同一模板换成相邻题型后是否还能用。
- 最后一轮做 [Challenge]，这些题用来确认概念已经能跨章节迁移。